# Lab Task 06

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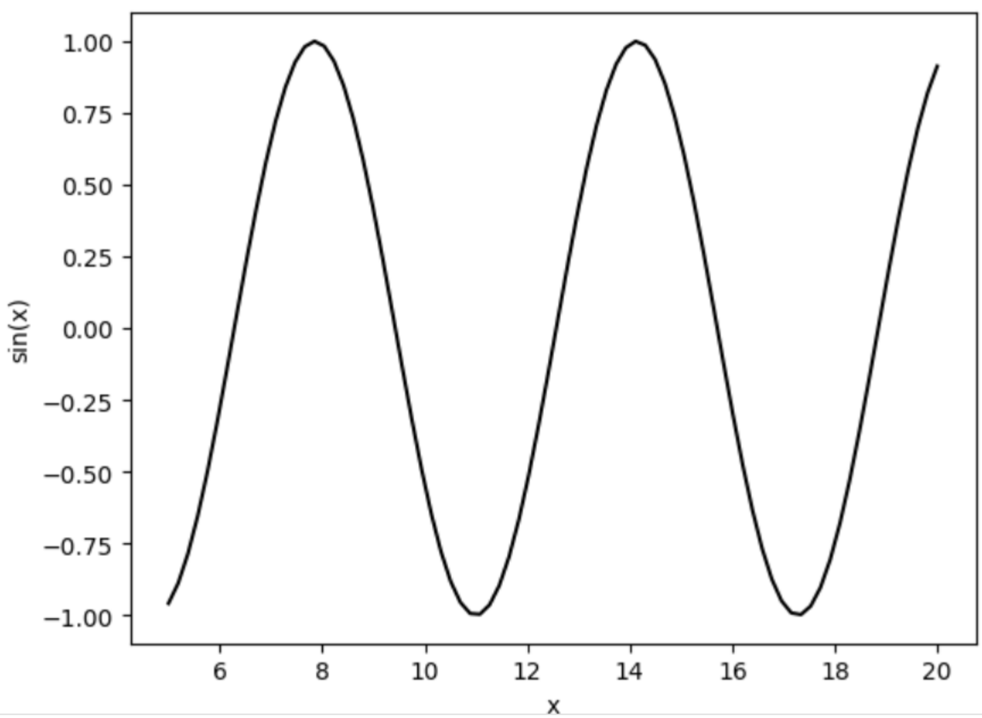
| **Submitted To** | **Ms. Amna Sajid** |
| **Roll No** | **FL-21565** |
| **Date** | **22-05-2025** |
| **Subject** | **AI** |

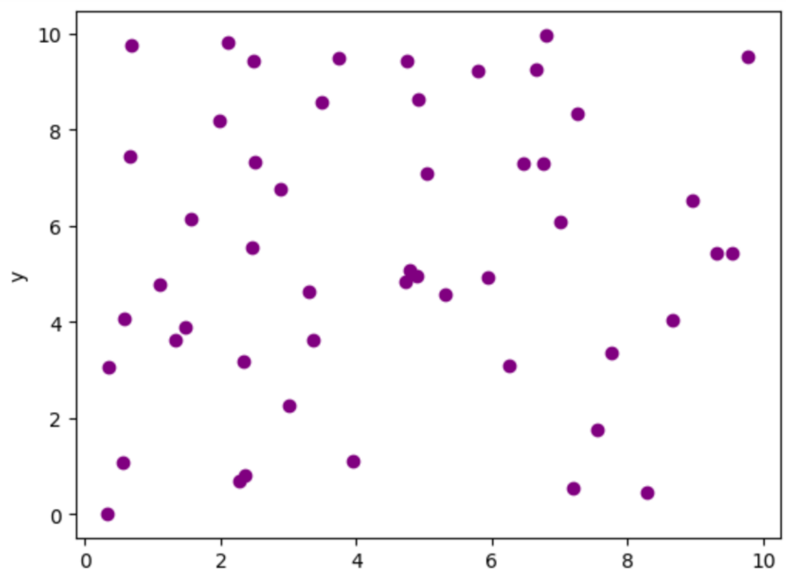
```python
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(5, 20, 80)
y = np.sin(x)

plt.plot(x, y, color="black") #use for line create
plt.xlabel('x')
plt.ylabel('sin(x)')
plt.show()
```

**Output**



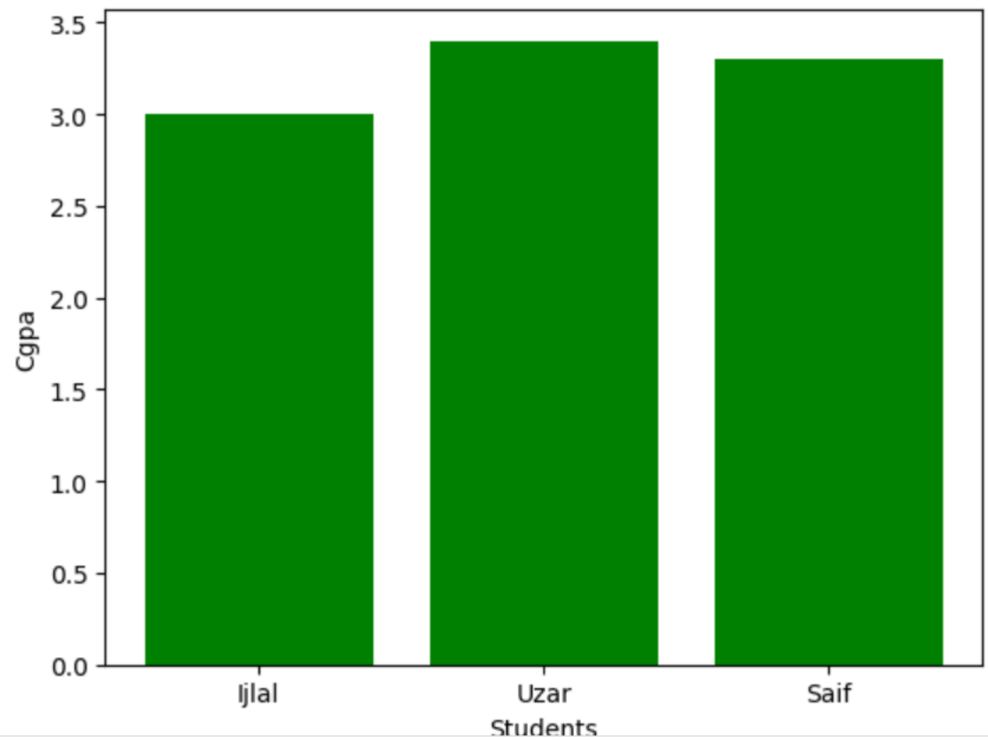
```python
x = np.random.uniform(0, 10, 50)
y = np.random.uniform(0, 10, 50)

plt.scatter(x, y, color = "purple")
plt.xlabel('x')
plt.ylabel('y')
plt.show()
```

**Output**



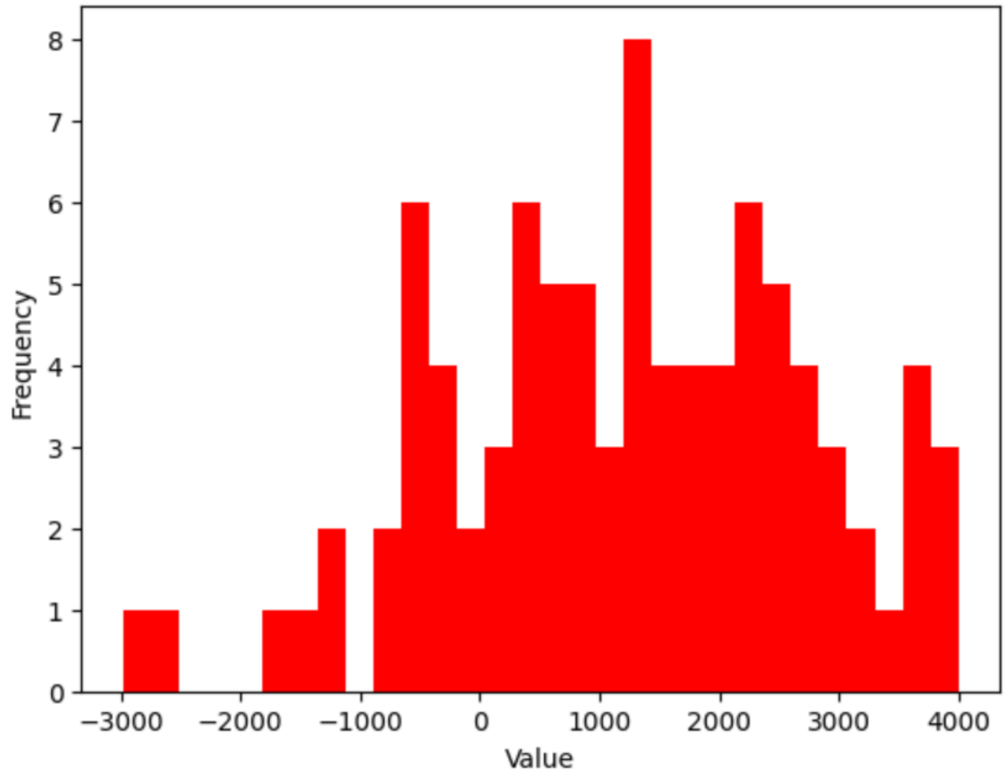

```python
Student = ['Ijlal', 'Uzar', 'Saif' ]
Cgpa = [3, 3.4, 3.3]

plt.bar(Student, Cgpa, color="green")
plt.xlabel('Students')
plt.ylabel('Cgpa')
plt.show()
```

## Output




```
data = np.random.normal(1000, 1500, 90)

plt.hist(data, bins=30, color="red")
plt.xlabel('Value')
plt.ylabel('Frequency')
plt.show()
```

## Output

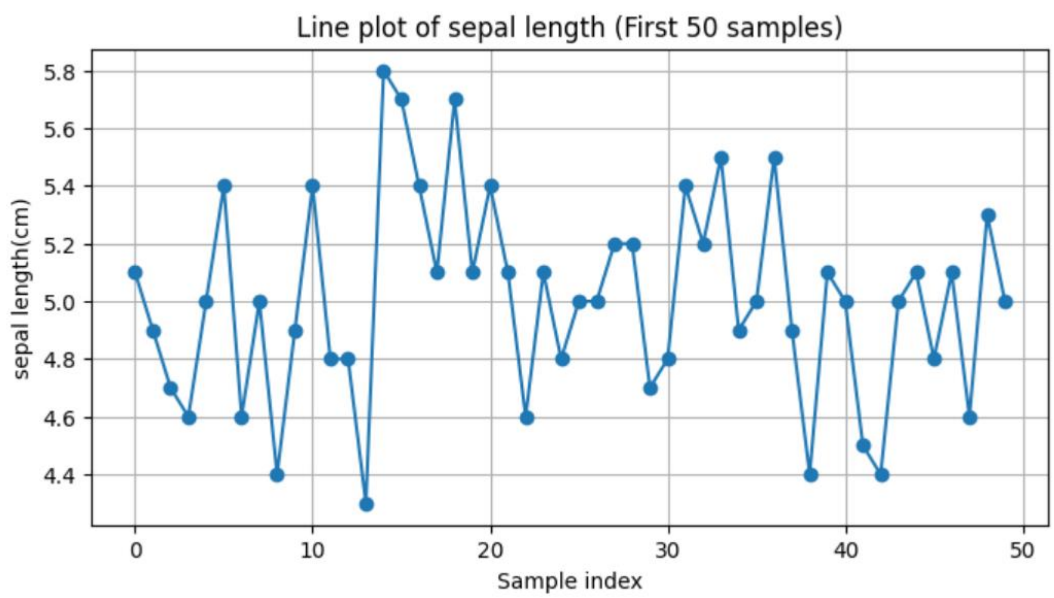



```
[ ] from sklearn.datasets import load_iris

    iris = load_iris()

    x = iris.data
    y = iris.target
    feature_names = iris.feature_names
    target_names = iris.target_names

    print("Featue names:", feature_names)
    print("Target names:", target_names)
    print("shape of data :", x.shape)
```

```
Featue names: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']
Target names: ['setosa' 'versicolor' 'virginica']
shape of data : (150, 4)
```

```python
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.datasets import load_iris
iris = load_iris()
df = pd.DataFrame(data=iris.data, columns=iris.feature_names)
df['species'] = iris.target
df['species_name'] = df['species'].apply(lambda x: iris.target_names[x])

plt.figure(figsize=(8, 4))
plt.plot(df['sepal length (cm)'][:50], marker='o') # Corrected markers to marker
plt.title("Line plot of sepal length (First 50 samples)")
plt.xlabel("Sample index") # Corrected xlable to xlabel
plt.ylabel("sepal length(cm)")
plt.grid(True)
plt.show()
```

## Output



```
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.datasets import load_iris
iris = load_iris()
df = pd.DataFrame(data=iris.data, columns=iris.feature_names)
df['species'] = iris.target
df['species_name'] = df['species'].apply(lambda x: iris.target_names[x])

plt.figure(figsize=(7, 4))
plt.hist(df['sepal width (cm)'], bins=15, color='lightgreen', edgecolor='black')
plt.title("Histogram of sepal width")
plt.xlabel("Sepal width (cm)")
plt.ylabel("Frequency")
plt.grid(True)
plt.show()
```

## Output